

Problem 2.9 *Characterize graphs G for which $rc(G) = src(G)$, or, give some sufficient conditions to guarantee $rc(G) = src(G)$.*

Recall the fact that if H is a connected spanning subgraph of a nontrivial (connected) graph G , then $rc(G) \leq rc(H)$. This fact is very useful to bounding the value of $rc(G)$ by giving bounds for its connected spanning subgraphs. We have noted that if, in addition, $diam(H) = 2$, then $src(G) \leq src(H)$. The authors of [12] naturally raised the following conjecture:

Conjecture 2.10 [12] *If H is a connected spanning subgraph of a nontrivial (connected) graph G , then $src(G) \leq src(H)$.*

Recently, this conjecture was disproved by Chakraborty, Fischer, Matsliah and Yuster [10]. They showed the following example: see Figure 2.1, here G is obtained from H by adding the edge $e = uv$, then H is a connected spanning subgraph of G . It is easy to show that there is a strong rainbow coloring of H which costs six colors, but the graph G costs at least seven colors to ensure its strong rainbow connection.

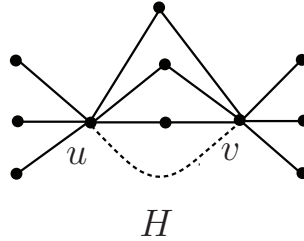


Figure 2.1 A counterexample to Conjecture 2.10.

Suppose that G contains two bridges $e = uv$ and $f = xy$. Then $G - e - f$ contains three components $G_i (1 \leq i \leq 3)$, where two of these components contain one of u, v, x and y and the third component contains two of these four vertices, say $u \in V(G_1)$, $x \in V(G_2)$ and $v, y \in V(G_3)$. If S is a set of k vertices contains u and x , then every tree whose vertex set contains S must also contain the edges e and f . This gives us a necessary condition for an edge-colored graph to be k -rainbow colored.

Observation 2.11 [15] *Let G be a connected graph of order n containing two bridges e and f . For each integer k with $2 \leq k \leq n$, every k -rainbow coloring of G must assign distinct colors to e and f .*

From Observation 2.11, we know that if G is rainbow connected under some edge-coloring, then any two bridges obtain distinct colors.

2.2 Upper bounds for rainbow connection number

We know that it is almost impossible to give the precise rainbow connection number of a given arbitrary graph, so we aim to give some nice bounds for it, especially sharp upper bounds.

In [9], Caro, Lev, Roditty, Tuza and Yuster investigated the extremal graph-theoretic behavior of rainbow connection number. Motivated by the fact that there are graphs with minimum degree 2 and with $rc(G) = n-3$ (just take two vertex-disjoint triangles and connect them by a path of length $n-5$), it is interesting to study the rainbow connection number of graphs with minimum degree at least 3 and they thought of the following question: is it true that minimum degree at least 3 guarantees $rc(G) \leq \alpha n$ where $\alpha < 1$ is independent of n ? This turns out to be true, and they proved:

Theorem 2.12 [9] *If G is a connected graph with n vertices and $\delta(G) \geq 3$, then $rc(G) < \frac{5}{6}n$.*

In the proof of Theorem 2.12, they first gave an upper bound for the rainbow connection number of 2-connected graphs (see Theorem 2.23), then from it, they next derived an upper bound for the rainbow connection number of connected bridgeless graphs (see Theorem 2.25).

The constant $5/6$ appearing in Theorem 2.12 is not optimal, but it probably cannot be replaced with a constant smaller than $\frac{3}{4}$, since there are 3-regular connected graphs with $rc(G) = diam(G) = \frac{3n-10}{4}$, and one of such graphs can be constructed as follows [53]: Take two vertex disjoint copies of the graph $K_5 - P_3$ and label the two vertices of degree 2 with w_1 and w_{2k+2} , where $k \geq 1$ is an integer. Next join w_1 and w_{2k+2} by a path of length $2k+1$ and label the vertices with $w_1, w_2, \dots, w_{2k+2}$. Now for $1 \leq i \leq k$ every edge $w_{2i}w_{2i+1}$ is replaced by a $K_4 - e$ and we identify the two vertices of degree 2 in $K_4 - e$ with w_{2i} and w_{2i+1} . The resulting graph G_{4k+10} is 3-regular, has order $n = 4k + 10$ and $rc(G_{4k+10}) = diam(G_{4k+10}) = 3k + 5 = \frac{3n-10}{4}$. Then Caro, Lev, Roditty, Tuza and Yuster conjectured:

Conjecture 2.13 [9] *If G is a connected graph with n vertices and $\delta(G) \geq 3$, then $rc(G) < \frac{3}{4}n$.*

Schiermeyer proved the conjecture in [53] by showing the following result:

Theorem 2.14 [53] *If G is a connected graph with n vertices and $\delta(G) \geq 3$, then $rc(G) < \frac{3n-1}{4}$.*

For 2-connected graphs Theorem 2.14 is true by Theorem 2.23. Hence it remains to prove it for graphs with connectivity 1. Schiermeyer extended the concept of rainbow connection number as follows: Additionally we require that any two edges of G have different colors whenever they belong to different blocks of G . The corresponding rainbow connection

number will be denoted by $rc^*(G)$. Then they derived Theorem 2.14 by first proving the following result: let G be a connected graph with n vertices, connectivity 1, and $\delta \geq 3$, then $rc^*(G) \leq \frac{3n-10}{4}$.

Not surprisingly, as the minimum degree increases, the graph would become more dense and therefore the rainbow connection number would decrease. Specifically, Caro, Lev, Roditty, Tuza and Yuster also proved the following upper bounds in term of minimum degree.

Theorem 2.15 [9] *If G is a connected graph with n vertices and minimum degree δ , then*

$$rc(G) \leq \min\left\{n\frac{\ln \delta}{\delta}(1 + o_\delta(1)), n\frac{4 \ln \delta + 3}{\delta}\right\}.$$

In the proof, they used the concept of a connected two-dominating set (A set of vertices S of G is called a *connected two-dominating set* if S induces a connected subgraph of G and, furthermore, each vertex outside of S has at least two neighbours in S) and the probabilistic method. They showed that in any case it cannot be improved below $\frac{3n}{\delta+1} - \frac{\delta+7}{\delta+1}$ as they constructed a connected n -vertex graph with minimum degree δ and this diameter: Take m copies of $K_{\delta+1}$, denoted by $X_1, \dots, X_m$ and label the vertices of X_i with $x_{i,1}, \dots, x_{i,\delta+1}$. Take two copies of $K_{\delta+2}$, denoted by X_0, X_{m+1} and similarly label their vertices. Now, connect $x_{i,2}$ with $x_{i+1,1}$ for $i = 0, \dots, m$ with an edge, and delete the edges $(x_{i,1}, x_{i,2})$ for $i = 0, \dots, m+1$. The obtained graph has $n = (m+2)(\delta+1) + 2$ vertices, and minimum degree δ (and maximum degree $\delta+1$). It is straightforward to verify that a shortest path from $x_{0,1}$ to $x_{m+1,2}$ has length $3m+5 = \frac{3n}{\delta+1} - \frac{\delta+7}{\delta+1}$.

This, naturally, raised the open problem of determining the true behavior of $rc(G)$ as a function of δ .

In [10], Chakraborty, Fischer, Matsliah and Yuster proved that any connected n -vertex graph with minimum degree $\Theta(n)$ has a bounded rainbow connection.

Theorem 2.16 [10] *For every $\epsilon > 0$ there is a constant $C = C(\epsilon)$ such that if G is a connected graph with n vertices and minimum degree at least ϵn , then $rc(G) \leq C$.*

The proof of Theorem 2.16 is based upon a modified degree-form version of Szemerédi Regularity Lemma (see [35] for a good survey on Regularity Lemma) that they proved.

The above lower bound construction suggests that the logarithmic factor in their upper bound may not be necessary and that, in fact $rc(G) \leq Cn/\delta$ where C is a universal constant. If true, notice that for graphs with a linear minimum degree ϵn , this implies that $rc(G)$ is at most C/ϵ . However, Theorem 2.16 does not even guarantee the weaker claim that $rc(G)$ is a constant. The constant $C = C(\epsilon)$ they obtained is a *tower function* in $1/\epsilon$ and in particular extremely far from being reciprocal to $1/\epsilon$.

Finally, Krivelevich and Yuster in [36] determined the behavior of $rc(G)$ as a function of $\delta(G)$ and resolved the above-mentioned open problem.